POST-ROUTE STATIC TIMING ANALYSIS

SKY130 implementation — report reconstructed from supplied terminal screenshots

Metric	Value from summary screenshot	Interpretation
Worst setup slack (WNS)	-1.762 ns	Setup timing fails; critical path needs optimization.
Total negative setup slack (TNS)	-69.502 ns	Aggregate setup violation reported by the tool.
Worst hold slack	+0.479 ns	Summary indicates hold timing passes.
Design area	5029 µm²	84% utilization.
Total power	7.34e-03 W (7.34 mW)	Combinational logic is the largest reported share.
Clock skew	0.00 ns setup skew	Reported source/target latency and CRPR sum to zero.

Executive summary

The supplied post-route report shows a significant setup-timing problem: WNS is –1.762 ns and TNS is –69.502 ns. The summary hold metric is positive (+0.479 ns), so the design-level summary indicates hold timing passes. Total reported power is 7.34 mW and design area is 5029 µm² at 84% utilization.

Important data note: some detailed path screenshots and the summary metrics do not reconcile exactly. For example, a detailed setup path shows –1.45 ns while the summary reports WNS –1.762 ns; one displayed min-path section shows a positive intermediate slack of 0.20 ns, while the summary reports worst hold slack +0.479 ns. This report preserves the values as shown and flags these differences rather than silently changing the source data.

1. Report order and timing interpretation

The report is arranged in the same logical section order as the terminal output: (1) setup max-delay path, (2) hold min-delay path, (3) worst setup slack, (4) worst hold slack, (5) WNS, (6) TNS, (7) clock skew, (8) slew/capacitance/fanout violations, (9) power, and (10) area. The provided screenshot for section 8 shows the heading but no listed violations beneath it.

Key detailed paths

Path	Startpoint → endpoint	Arrival time	Displayed slack / note
Setup, register-to-register	dpath.b_reg.out[0] → dpath.a_reg.out[9]	2.55 ns	–1.45 ns (VIOLATED)
Hold, register-to-register	ctrl.state.out[1] → ctrl.state.out[1]	0.68 ns	Displayed min-path arithmetic is inconsistent in screenshot; su
Setup, register-to-output	dpath.b_reg.out[0] → resp_msg[3]	2.64 ns	Output path detail shown; endpoint requirement lines are not f
Min, register-to-output	ctrl.state.out[1] → resp_val	0.84 ns	+1.06 ns (MET) in screenshot.

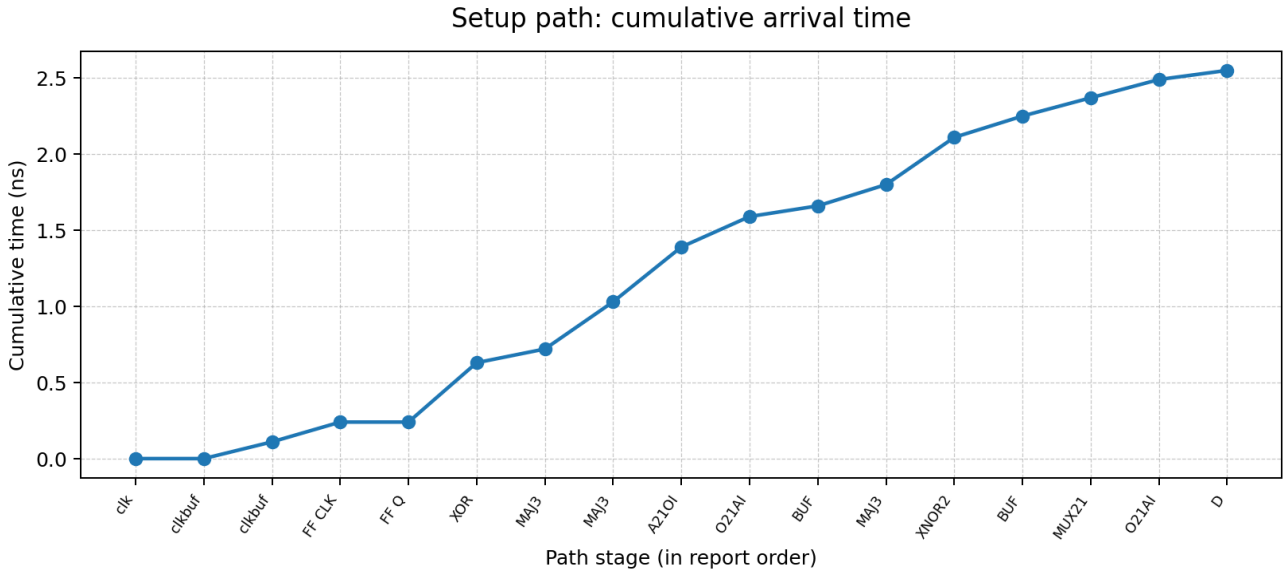
Timing terms

Setup: data must arrive before the capture edge minus setup time. Negative slack means a setup violation. **Hold:** data must not arrive too soon after the capture edge plus hold time. Positive slack means the checked path meets that requirement. **WNS:** worst negative slack across setup paths. **TNS:** total setup violation magnitude as reported by the tool (sign convention retained from screenshot).

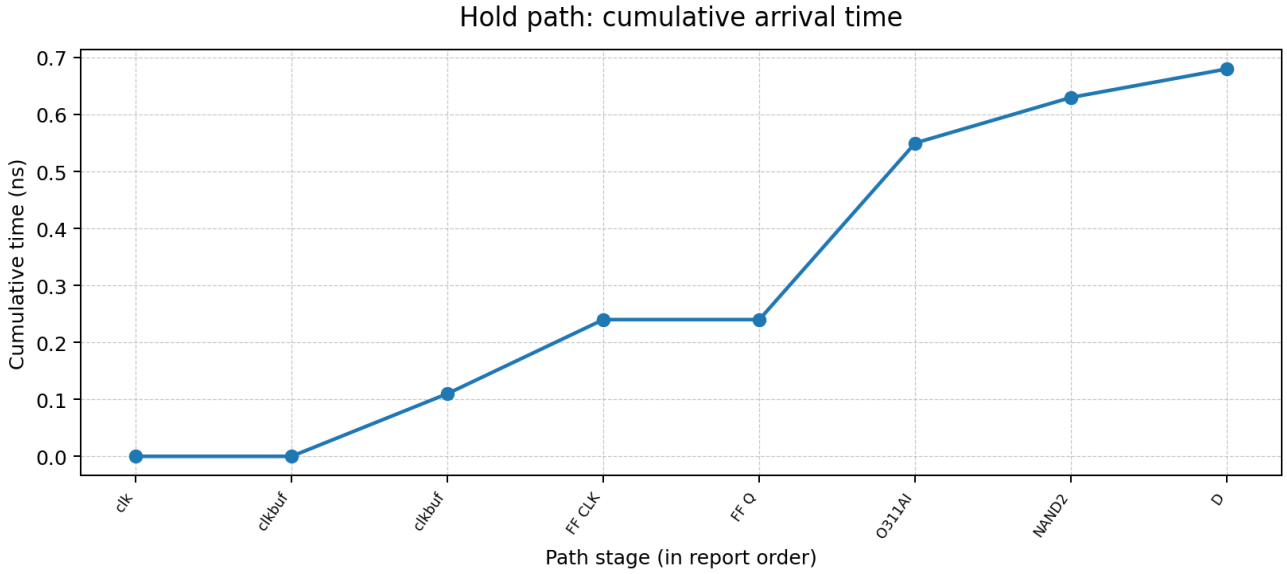
2. Cumulative timing-path graphs

The line plots below use the cumulative “Time” column transcribed from the supplied path reports. The x-axis follows the order of path stages shown in each terminal report; stage labels are shortened to keep the graphs legible. Values are in nanoseconds, as implied by the report’s timing values.

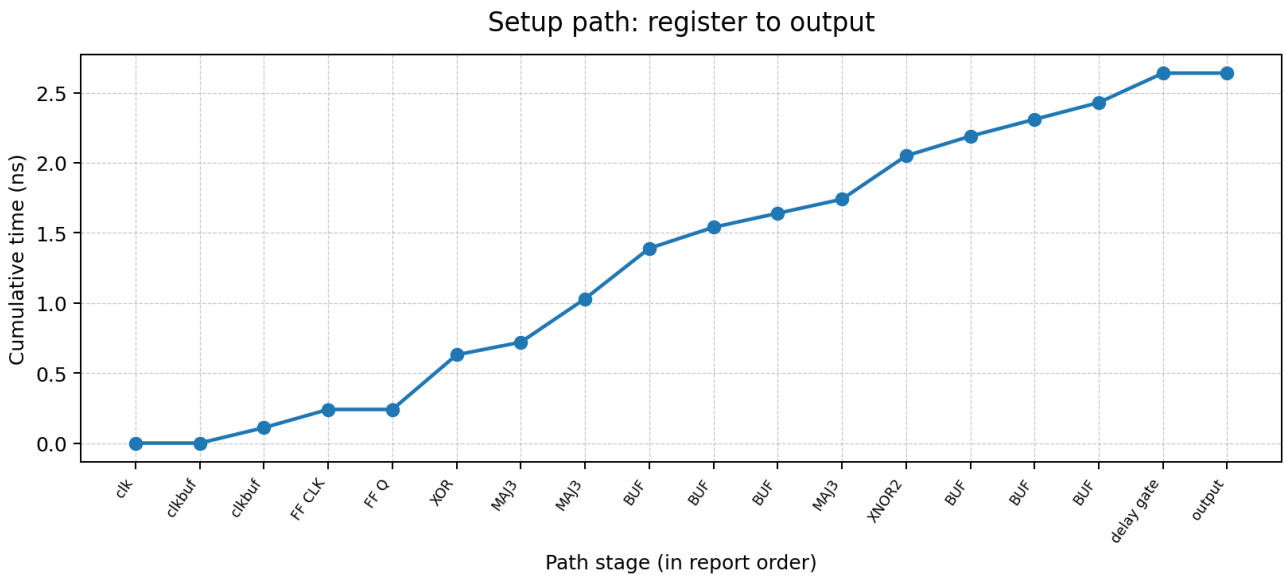
2.1 Setup: register-to-register path



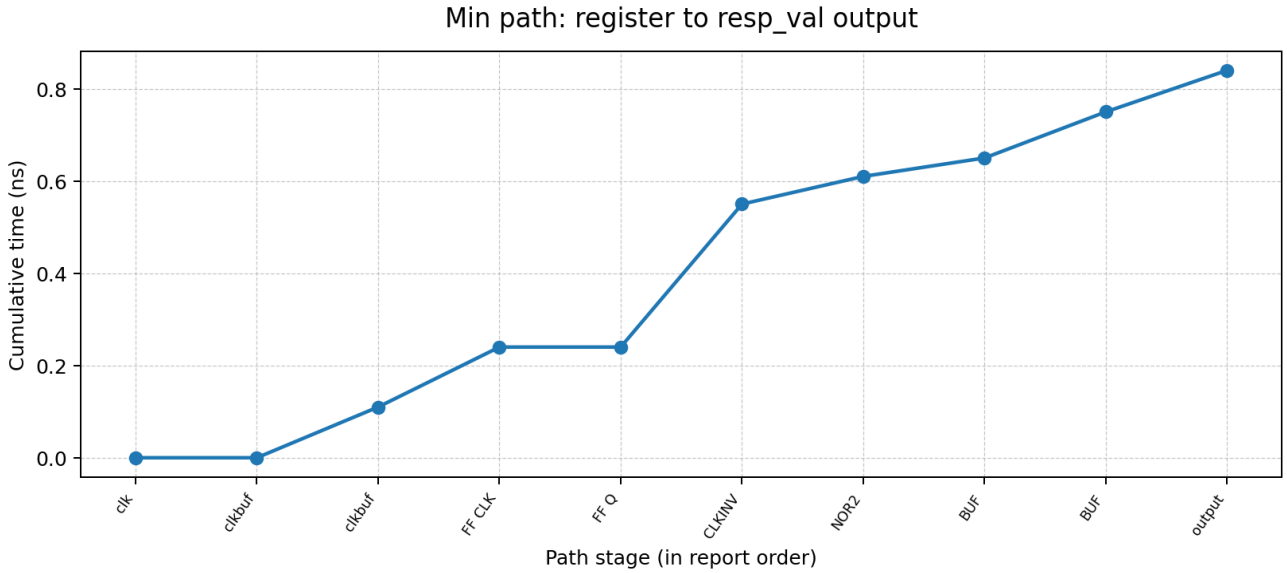
2.2 Hold: register-to-register path



2.3 Setup: register-to-output path

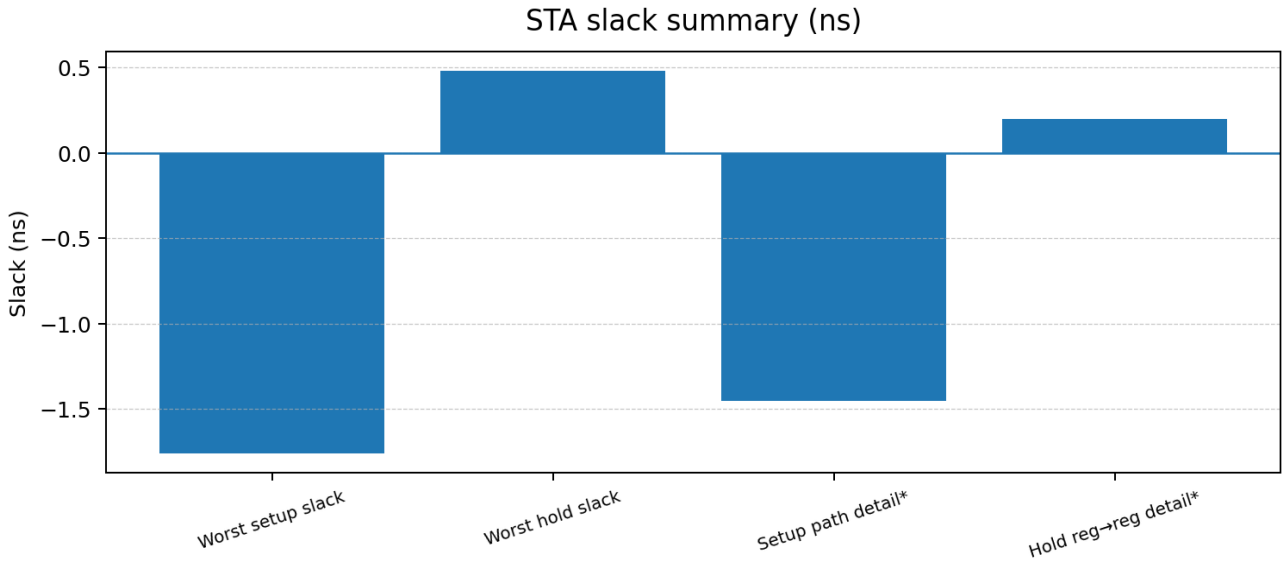


2.4 Min path: register-to-resp_val output



3. Slack, power, and area

The slack comparison below juxtaposes summary values with selected detailed-path values. Detailed-path values are included for context and are not substitutes for the design-level WNS or worst hold slack.



Power breakdown (transcribed)

Group	Internal power (W)	Switching power (W)	Leakage power (W)	Total power (W)	Share
Sequential	1.65e-03	1.06e-04	3.51e-10	1.75e-03	23.9%
Combinational	3.51e-03	1.36e-03	1.87e-09	4.87e-03	66.4%
Clock	4.28e-04	2.88e-04	4.77e-11	7.15e-04	9.8%
Macro	0	0	0	0	0.0%
Pad	0	0	0	0	0.0%
Total	**5.58e-03**	**1.75e-03**	**2.27e-09**	**7.34e-03**	**100.0%**

Area: 5029 µm², 84% utilization. **Slew/cap/fanout:** section heading is present in the screenshot, but no violations are listed below it. **Clock skew:** core_clock setup skew reported as 0.00 ns; the virtual-clock section says no launch/capture paths were found.

Appendix A. Supplied terminal screenshots

Images are included in the order needed to follow the report: setup path and continuation; hold path and continuation; setup output path; hold/output path details; summary slack/skew; power and area.

A1. Setup timing — register-to-register path

```
=========== 1. SETUP TIMING (max-delay paths) ===========
Startpoint: dpath.b_reg.out[0]$_DFFE_PP_
            (rising edge-triggered flip-flop clocked by core_clock)
Endpoint: dpath.a_reg.out[9]$_DFFE_PP_
          (rising edge-triggered flip-flop clocked by core_clock)
Path Group: core_clock
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clock (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    0.24 ^ dpath.b_reg.out[0]$_DFFE_PP_/CLK (sky130_fd_sc_hd__edfxtp_1)
   0.38    0.63 ^ dpath.b_reg.out[0]$_DFFE_PP_/Q (sky130_fd_sc_hd__edfxtp_1)
   0.09    0.72 v _120_/_113_/X (sky130_fd_sc_hd__xor2_2)
   0.31    1.03 v _120_/_115_/X (sky130_fd_sc_hd__maj3_2)
   0.36    1.39 v _120_/_117_/X (sky130_fd_sc_hd__maj3_2)
   0.20    1.59 ^ _120_/_119_/Y (sky130_fd_sc_hd__a21oi_4)
   0.07    1.66 v _120_/_120_/Y (sky130_fd_sc_hd__o21ai_2)
   0.14    1.80 v _120_/place165/X (sky130_fd_sc_hd__buf_4)
   0.31    2.11 v _120_/_173_/X (sky130_fd_sc_hd__maj3_2)
   0.14    2.25 v _120_/_174_/Y (sky130_fd_sc_hd__xnor2_2)
   0.12    2.37 v _120_/rebuffer278/X (sky130_fd_sc_hd__buf_4)
   0.11    2.49 ^ _142_/Y (sky130_fd_sc_hd__mux2i_1)
   0.06    2.55 v _144_/Y (sky130_fd_sc_hd__o21ai_1)
   0.00    2.55 v dpath.a_reg.out[9]$_DFFE_PP_/D (sky130_fd_sc_hd__edfxtp_1)
           2.55   data arrival time

   1.10    1.10   clock core_clock (rise edge)
```

A2. Setup timing — continuation and slack

```
   0.00    2.55 v dpath.a_reg.out[9]$_DFFE_PP_/D (sky130_fd_sc_hd__edfxtp_1)
           2.55   data arrival time

   1.10    1.10   clock core_clock (rise edge)
   0.00    1.10   clock source latency
   0.00    1.10 ^ clk (in)
   0.11    1.21 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    1.34 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    1.34 ^ dpath.a_reg.out[9]$_DFFE_PP_/CLK (sky130_fd_sc_hd__edfxtp_1)
   0.00    1.34   clock reconvergence pessimism
  -0.24    1.10   library setup time
           1.10   data required time
-----------------------------------------------------------
           1.10   data required time
          -2.55   data arrival time
-----------------------------------------------------------
          -1.45   slack (VIOLATED)
```

A3. Hold timing — register-to-register path

```
=========== 2. HOLD TIMING (min-delay paths) ===========
Startpoint: ctrl.state.out[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by core_clock)
Endpoint: ctrl.state.out[1]$_DFF_P_
          (rising edge-triggered flip-flop clocked by core_clock)
Path Group: core_clock
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clock (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    0.24 ^ ctrl.state.out[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.30    0.55 v ctrl.state.out[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.08    0.63 ^ _202_/Y (sky130_fd_sc_hd__o311ai_0)
   0.05    0.68 v _204_/Y (sky130_fd_sc_hd__nand2_1)
   0.00    0.68 v ctrl.state.out[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.68   data arrival time
```

A4. Hold timing — continuation and slack

```
   1.10    1.10   clock vclk_core_clock (rise edge)
   0.00    1.10   clock network delay (ideal)
   0.00    1.10   clock reconvergence pessimism
  -0.22    0.88   output external delay
           0.88   data required time
-----------------------------------------------------------
           0.88   data required time
          -2.64   data arrival time
-----------------------------------------------------------
          -1.76   slack (VIOLATED)
```

A5. Setup timing — register-to-output path

```
Startpoint: dpath.b_reg.out[0]$_DFFE_PP_
            (rising edge-triggered flip-flop clocked by core_clock)
Endpoint: resp_msg[3] (output port clocked by vclk_core_clock)
Path Group: vclk_core_clock
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clock (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    0.24 ^ dpath.b_reg.out[0]$_DFFE_PP_/CLK (sky130_fd_sc_hd__edfxtp_1)
   0.38    0.63 ^ dpath.b_reg.out[0]$_DFFE_PP_/Q (sky130_fd_sc_hd__edfxtp_1)
   0.09    0.72 v _120_/_113_/X (sky130_fd_sc_hd__xor2_2)
   0.31    1.03 v _120_/_115_/X (sky130_fd_sc_hd__maj3_2)
   0.36    1.39 v _120_/_117_/X (sky130_fd_sc_hd__maj3_2)
   0.15    1.54 v _120_/rebuffer281/X (sky130_fd_sc_hd__buf_4)
   0.10    1.64 v _120_/rebuffer232/X (sky130_fd_sc_hd__buf_6)
   0.10    1.74 v _120_/place176/X (sky130_fd_sc_hd__buf_6)
   0.31    2.05 v _120_/_162_/X (sky130_fd_sc_hd__maj3_2)
   0.14    2.19 v _120_/_163_/Y (sky130_fd_sc_hd__xnor2_2)
   0.12    2.31 v _120_/place164/X (sky130_fd_sc_hd__buf_4)
   0.11    2.43 v _120_/place74/X (sky130_fd_sc_hd__buf_4)
   0.22    2.64 v output48/X (sky130_fd_sc_hd__dlygate4sd2_1)
   0.00    2.64 v resp_msg[3] (out)
           2.64   data arrival time

   1.10    1.10   clock vclk_core_clock (rise edge)
   0.00    1.10   clock network delay (ideal)
   0.00    1.10   clock reconvergence pessimism
```

A6. Hold timing — register-to-output / slack detail

```
  0.00    0.00   clock core_clock (rise edge)
  0.00    0.00   clock source latency
  0.00    0.00 ^ clk (in)
  0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
  0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
  0.00    0.24 ^ ctrl.state.out[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  0.00    0.24   clock reconvergence pessimism
 -0.04    0.20   library hold time
          0.20   data required time
-----------------------------------------------------------
          0.20   data required time
         -0.68   data arrival time
-----------------------------------------------------------
          0.48   slack (MET)
```

A7. Hold/output timing path details

```
Startpoint: ctrl.state.out[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by core_clock)
Endpoint: resp_val (output port clocked by vclk_core_clock)
Path Group: vclk_core_clock
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock core_clock (rise edge)
   0.00    0.00   clock source latency
   0.00    0.00 ^ clk (in)
   0.11    0.11 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.13    0.24 ^ clkbuf_2_0__f_clk/X (sky130_fd_sc_hd__clkbuf_8)
   0.00    0.24 ^ ctrl.state.out[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.30    0.55 v ctrl.state.out[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.06    0.61 ^ _123_/Y (sky130_fd_sc_hd__clkinv_1)
   0.04    0.65 v _205_/Y (sky130_fd_sc_hd__nor2_4)
   0.10    0.75 v place180/X (sky130_fd_sc_hd__buf_6)
   0.09    0.84 v output55/X (sky130_fd_sc_hd__buf_6)
   0.00    0.84 v resp_val (out)
           0.84   data arrival time

   0.00    0.00   clock vclk_core_clock (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
  -0.22   -0.22   output external delay
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.84   data arrival time
---------------------------------------------------------
           1.06   slack (MET)
```

A8. Worst slack, WNS/TNS, and clock skew summary

```
========== 8. SLEW / CAP / FANOUT VIOLATORS ==========

========== 9. POWER ==========
Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             1.65e-03   1.06e-04   3.51e-10   1.75e-03  23.9%
Combinational          3.51e-03   1.36e-03   1.87e-09   4.87e-03  66.4%
Clock                  4.28e-04   2.88e-04   4.77e-11   7.15e-04   9.8%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  5.58e-03   1.75e-03   2.27e-09   7.34e-03 100.0%
                          76.1%      23.9%       0.0%

========== 10. AREA ==========
Design area 5029 um^2 84% utilization.

Post-route STA report commands completed.
```

A9. Slew/cap/fanout, power, and area summary

```
=========== 3. WORST SETUP SLACK ===========
worst slack max -1.762

=========== 4. WORST HOLD SLACK ===========
worst slack min 0.479

=========== 5. WNS (setup) ===========
wns max -1.762

=========== 6. TNS (setup) ===========
tns max -69.502

=========== 7. CLOCK SKEW ===========
Clock core_clock
   0.24 source latency ctrl.state.out[0]$_DFF_P_/CLK ^
  -0.24 target latency dpath.a_reg.out[11]$_DFFE_PP_/CLK ^
   0.00 CRPR
--------------
   0.00 setup skew

Clock vclk_core_clock
No launch/capture paths found.
```